iamneo

Problem Solving - Flowchart

What is a Flowchart?

A flowchart is a graphical representation of a process, algorithm, or workflow that uses standardized symbols connected by arrows to show the sequence of steps.

- Visual representation of step-by-step solution
- Uses standardized symbols and connectors
- Shows logical flow from start to end
- Universal language for problem-solving

History & Importance

History

- Introduced by Frank Gilbreth in 1921
- Popularized in computer programming in 1940s
- Standardized by ANSI in 1960s

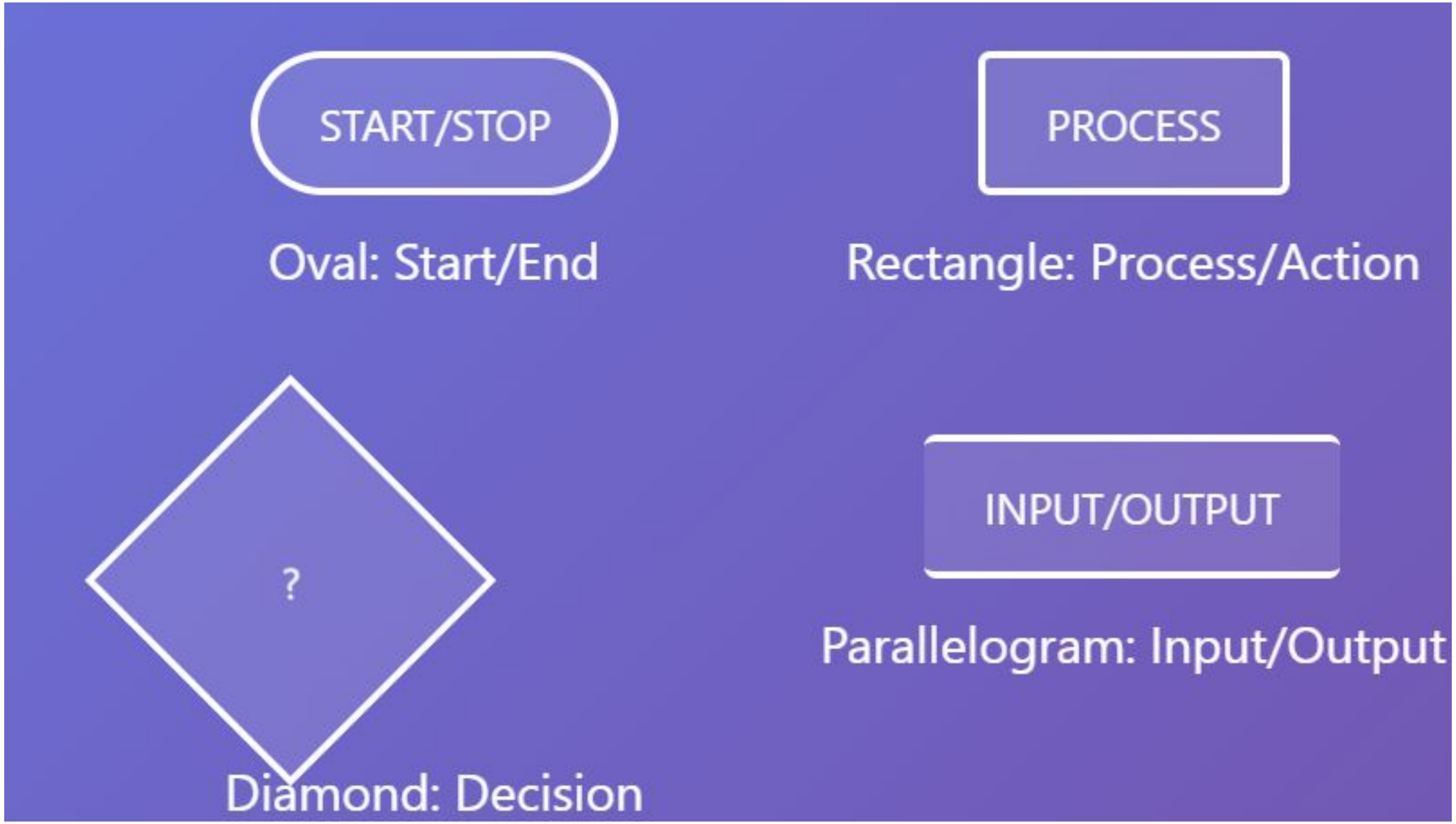
Importance

- Simplifies complex problems
- Improves communication
- Helps in debugging
- Documentation tool

Basic Flowchart Symbols

Symbol	Name	Function
	Start/end	An oval represents a start or end point.
	Arrows	A line is a connector that shows relationships between the representative shapes.
	Input/Output	A parallelogram represents input or ouptut.
	Process	A rectangle represents a process.
	Decision	A diamond indicates a decision.

Basic Flowchart Symbols

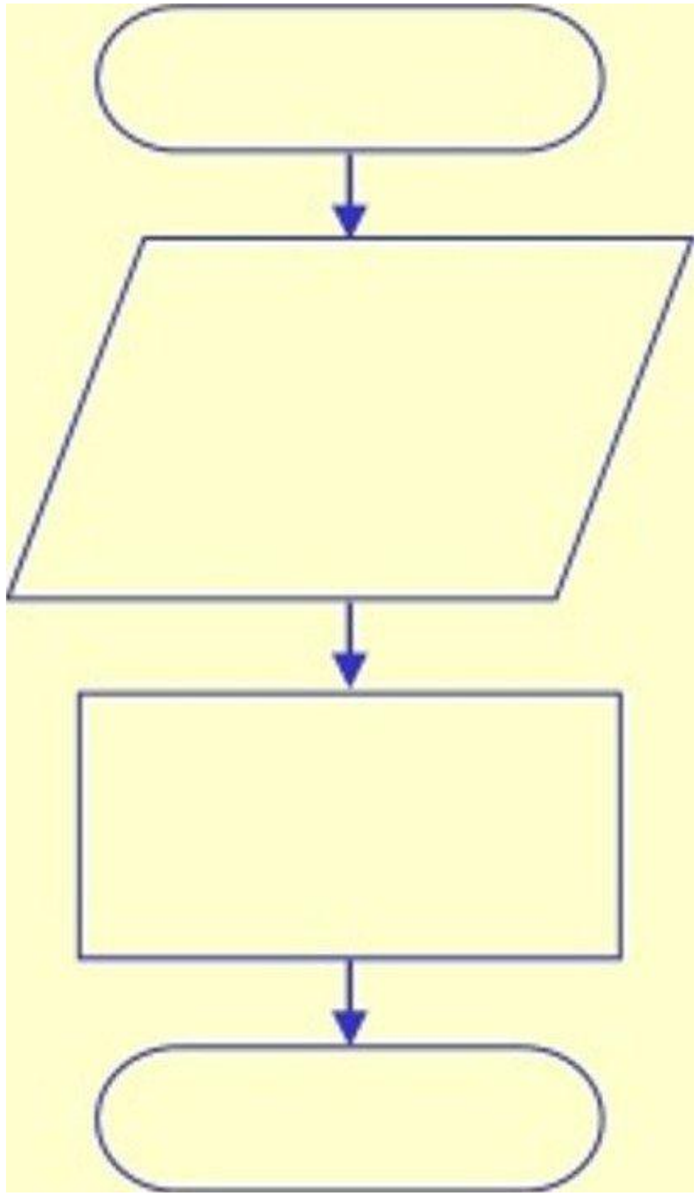


Four Flowchart Structures

- Sequence
- Decision
- Repetition
- Case

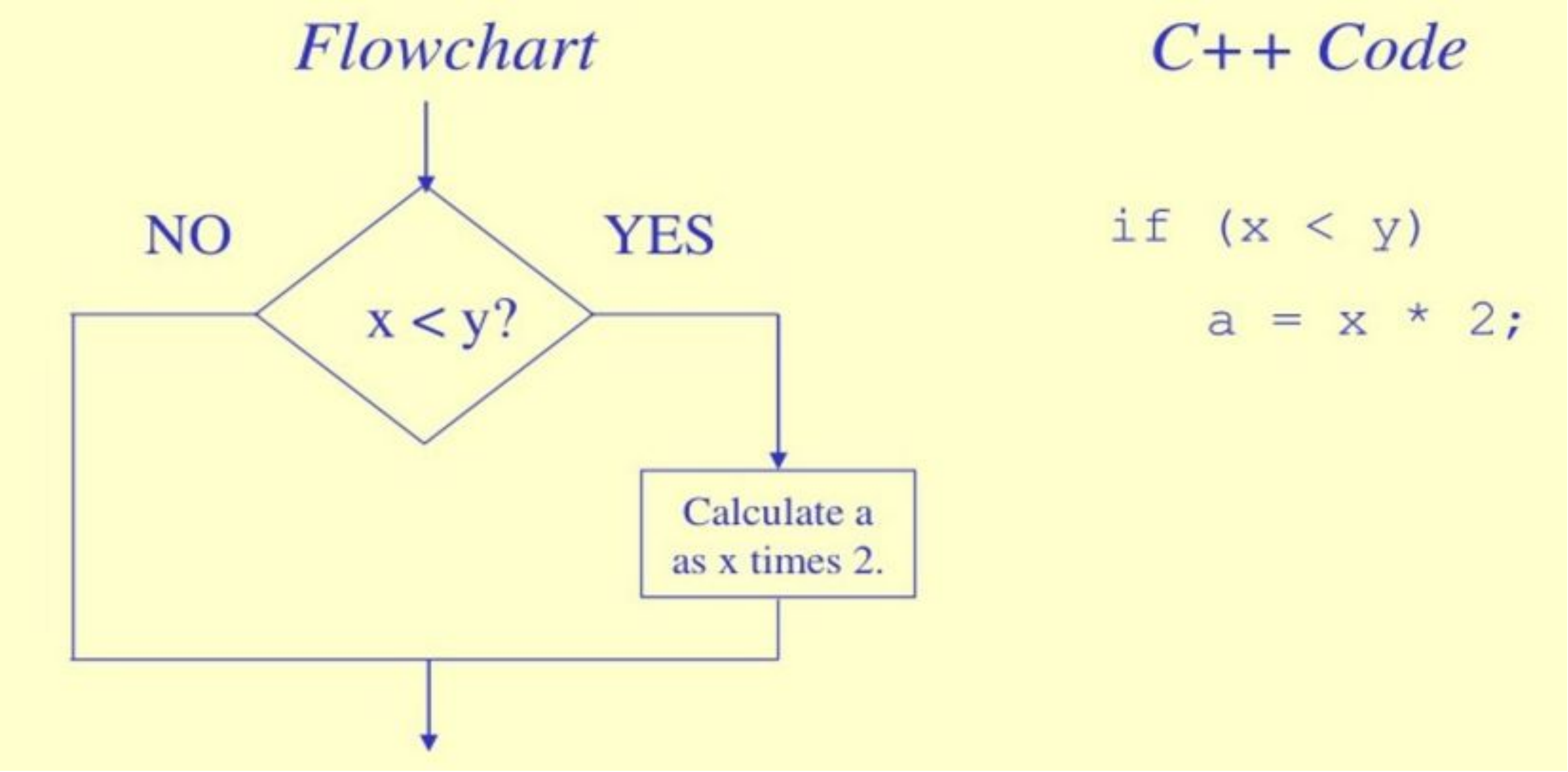
Sequence Structure

- A series of actions are performed in sequence
- The pay-calculating example was a sequence flowchart.



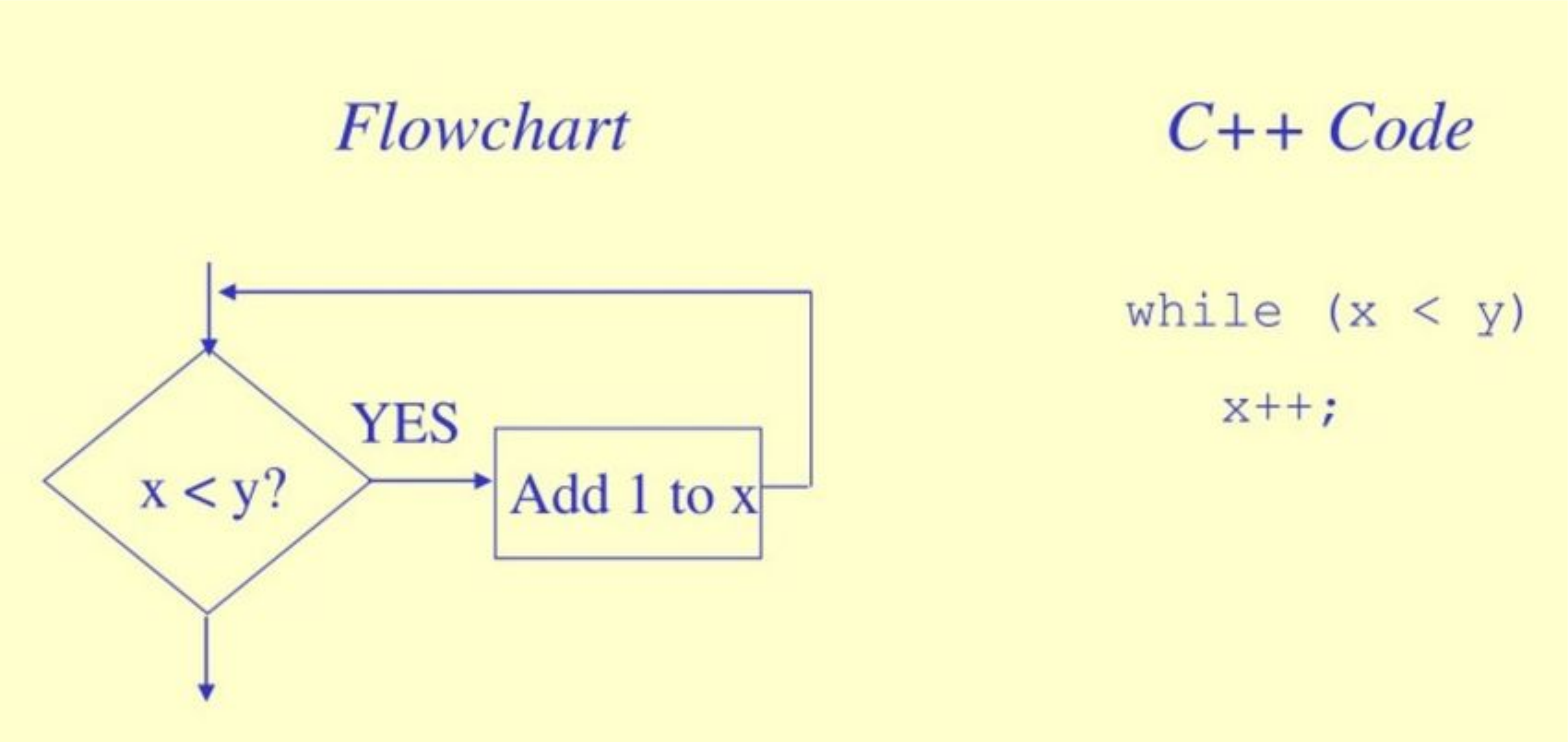
Decision Structure

- The flowchart segment below shows a decision structure with only one a ction to perform. It is expressed as an if statement in C++ code.



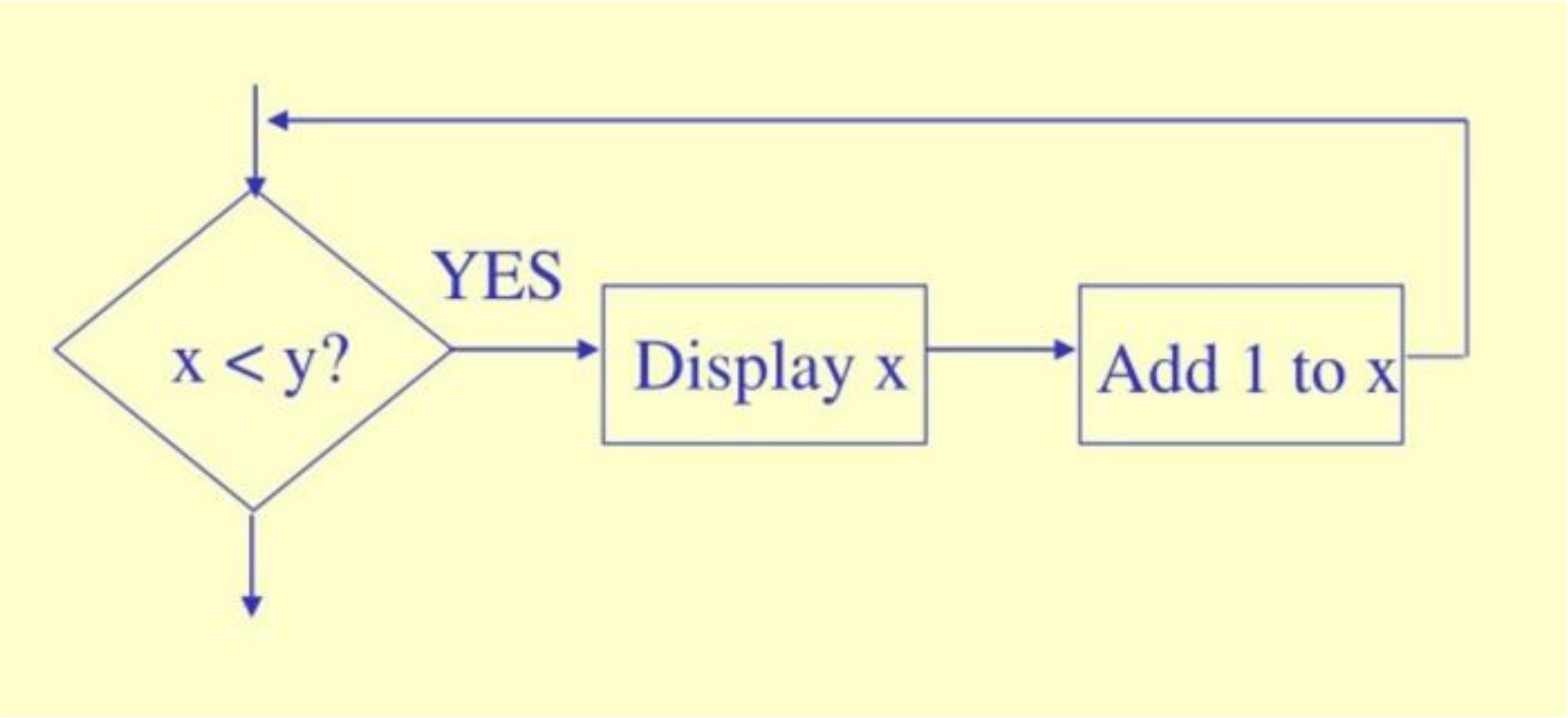
Repetition Structure

The flowchart segment below shows a repetition structure expressed in C++ as a while loop

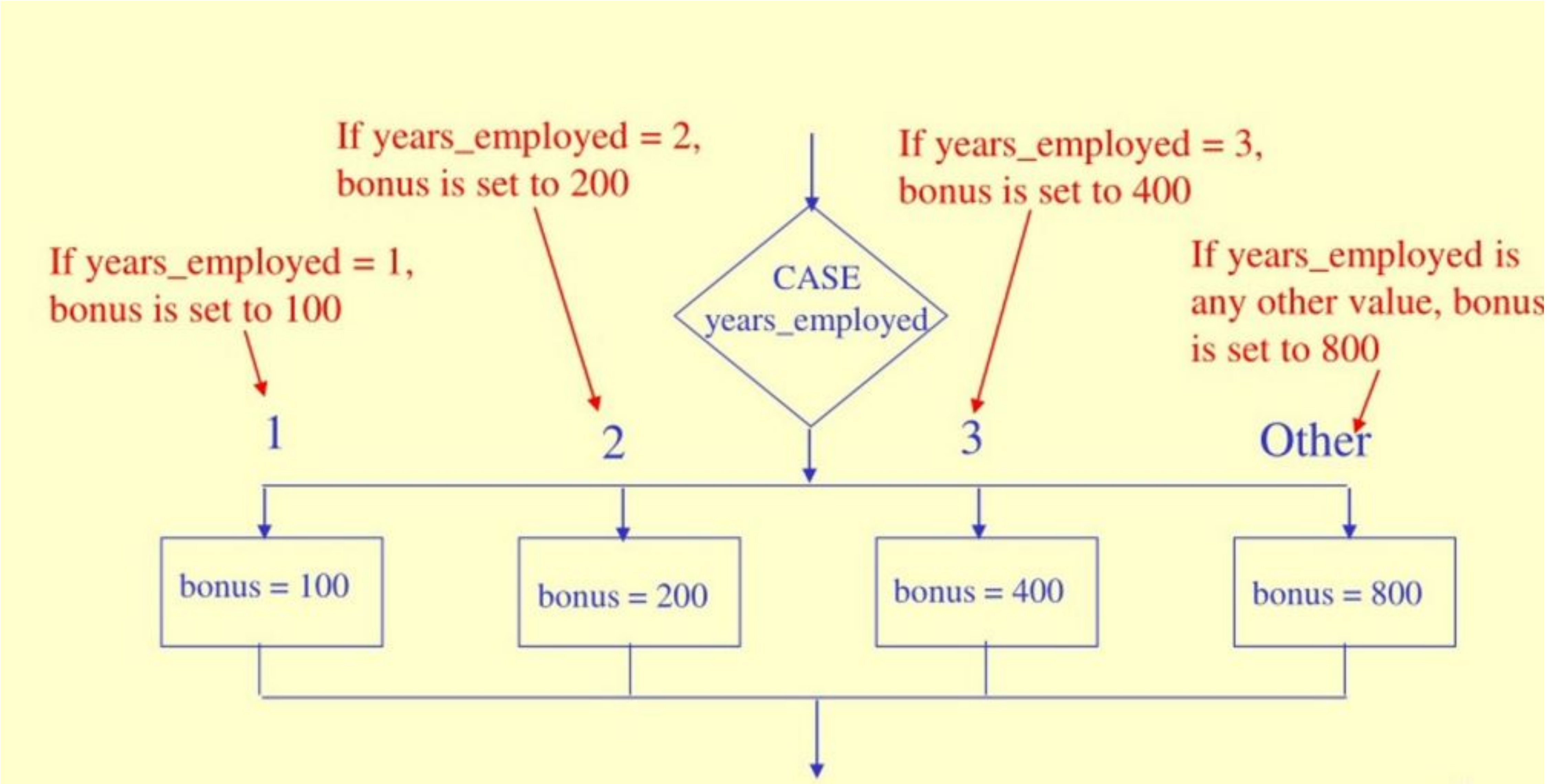


Controlling a Repetition Structure

By adding an action within the repetition that changes the value of x.

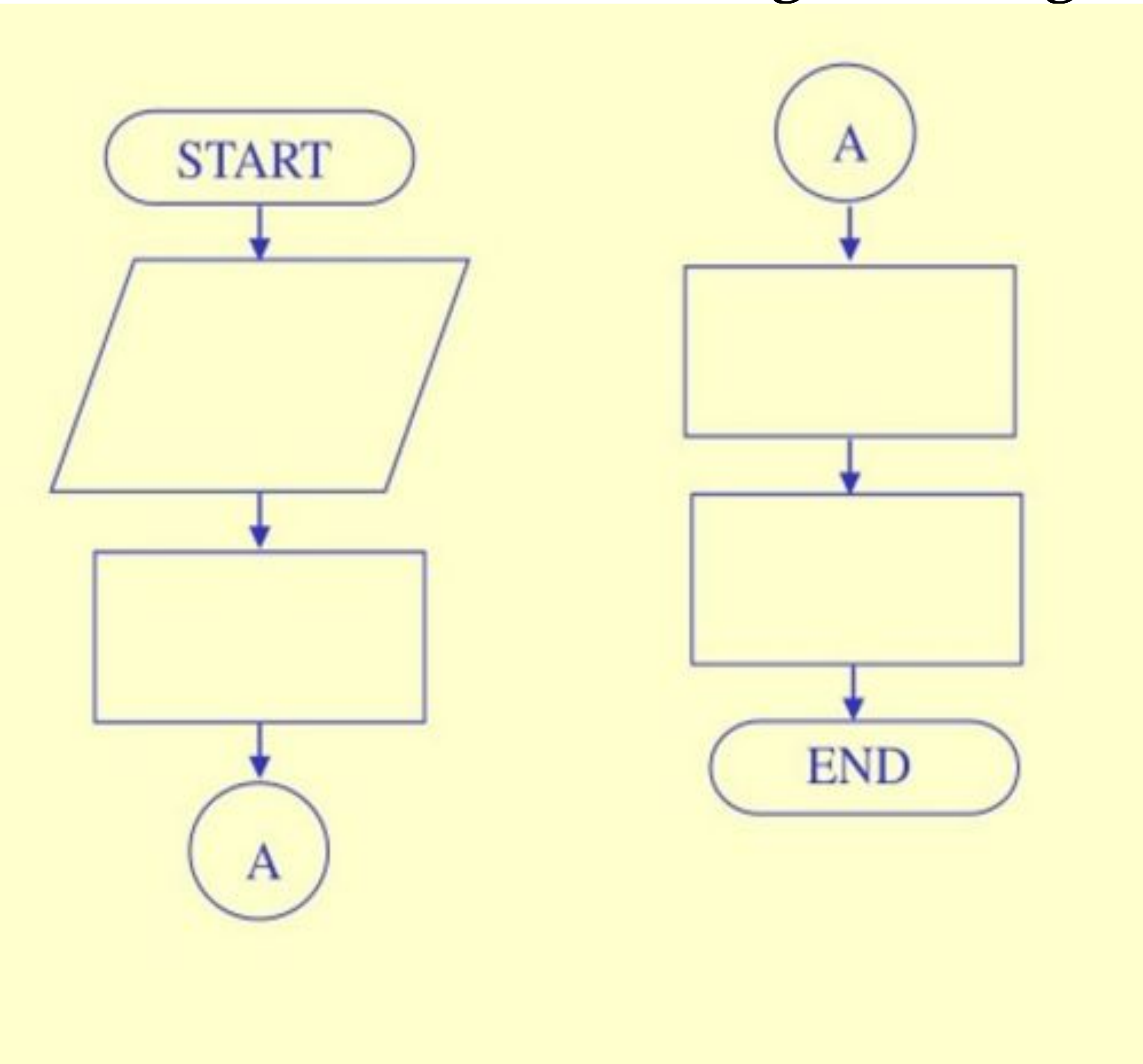


Case Structure



Connectors

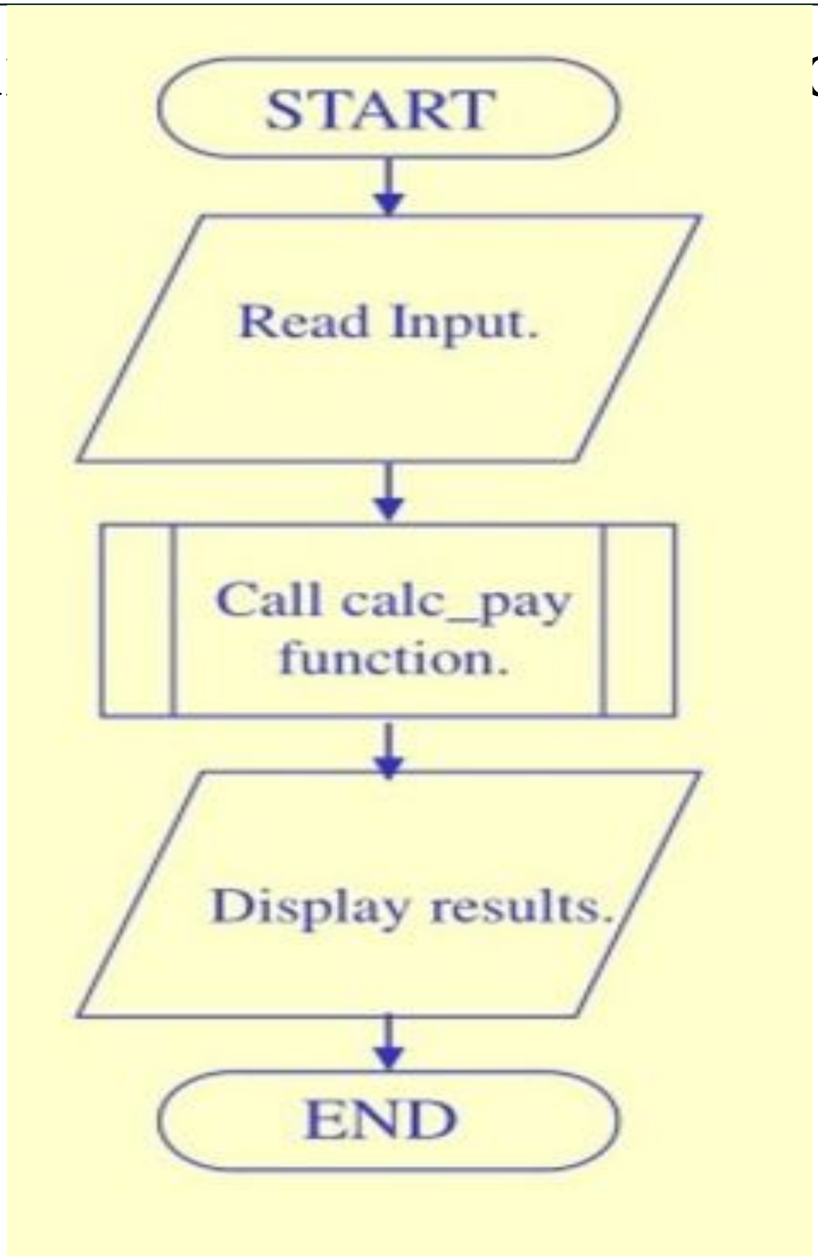
•The “A” connector indicates that the second flowchart segment begins where the first segment ends.



Modules

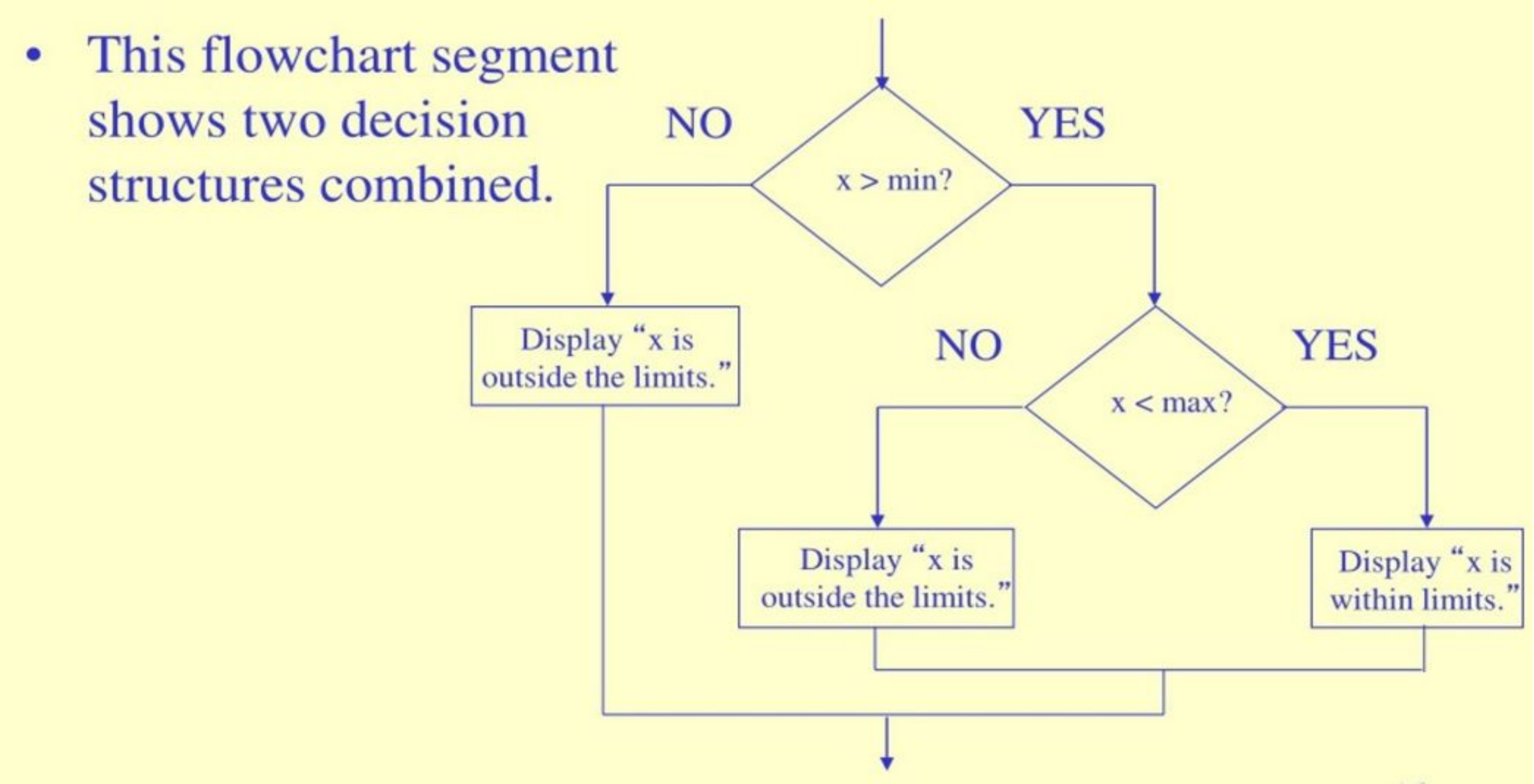
- The position of the module symbol ind odule is exe

- A separate flowchart can be constructed for the module.



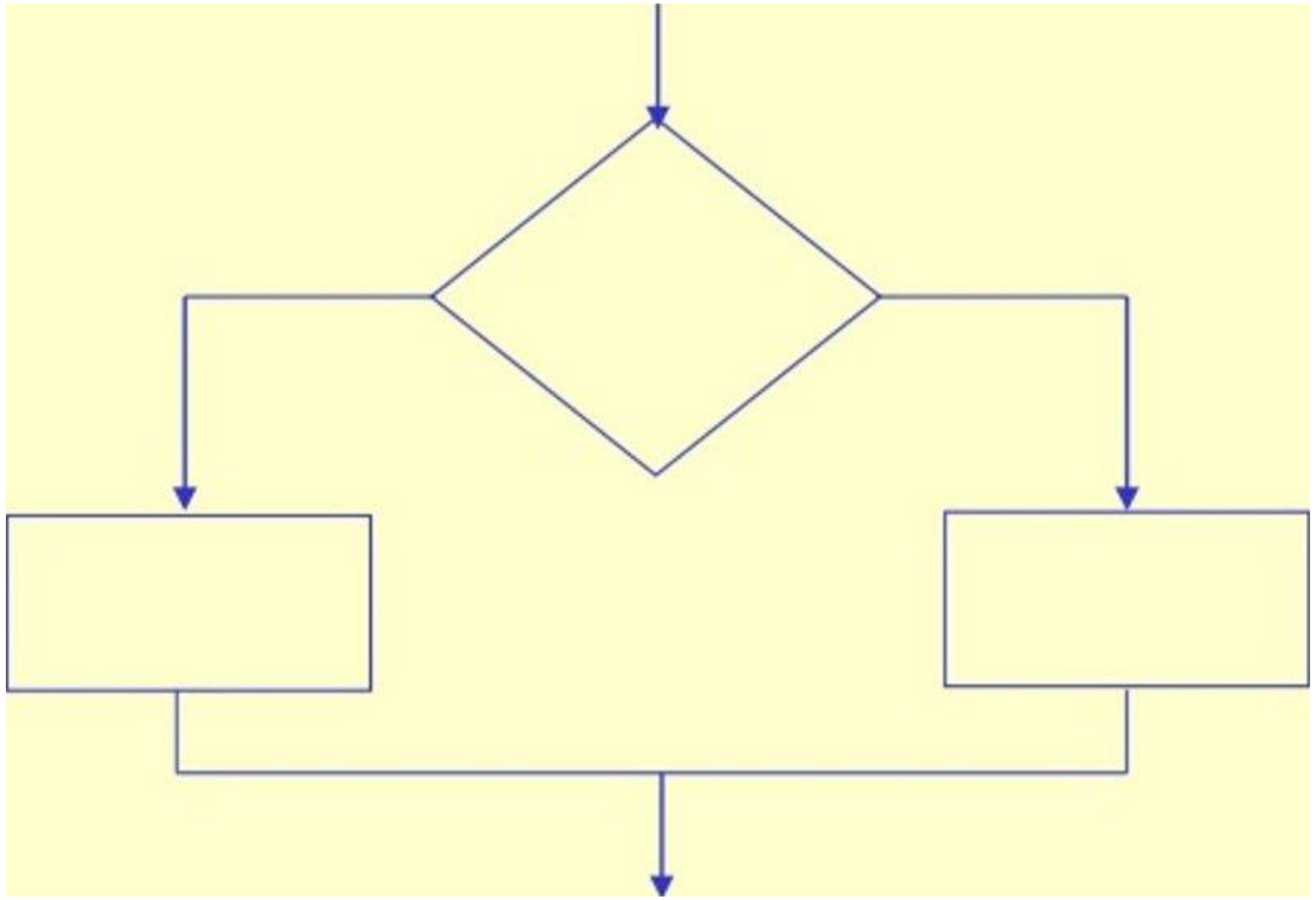
Combining Structures

This flowchart segment shows two decision structures combined.



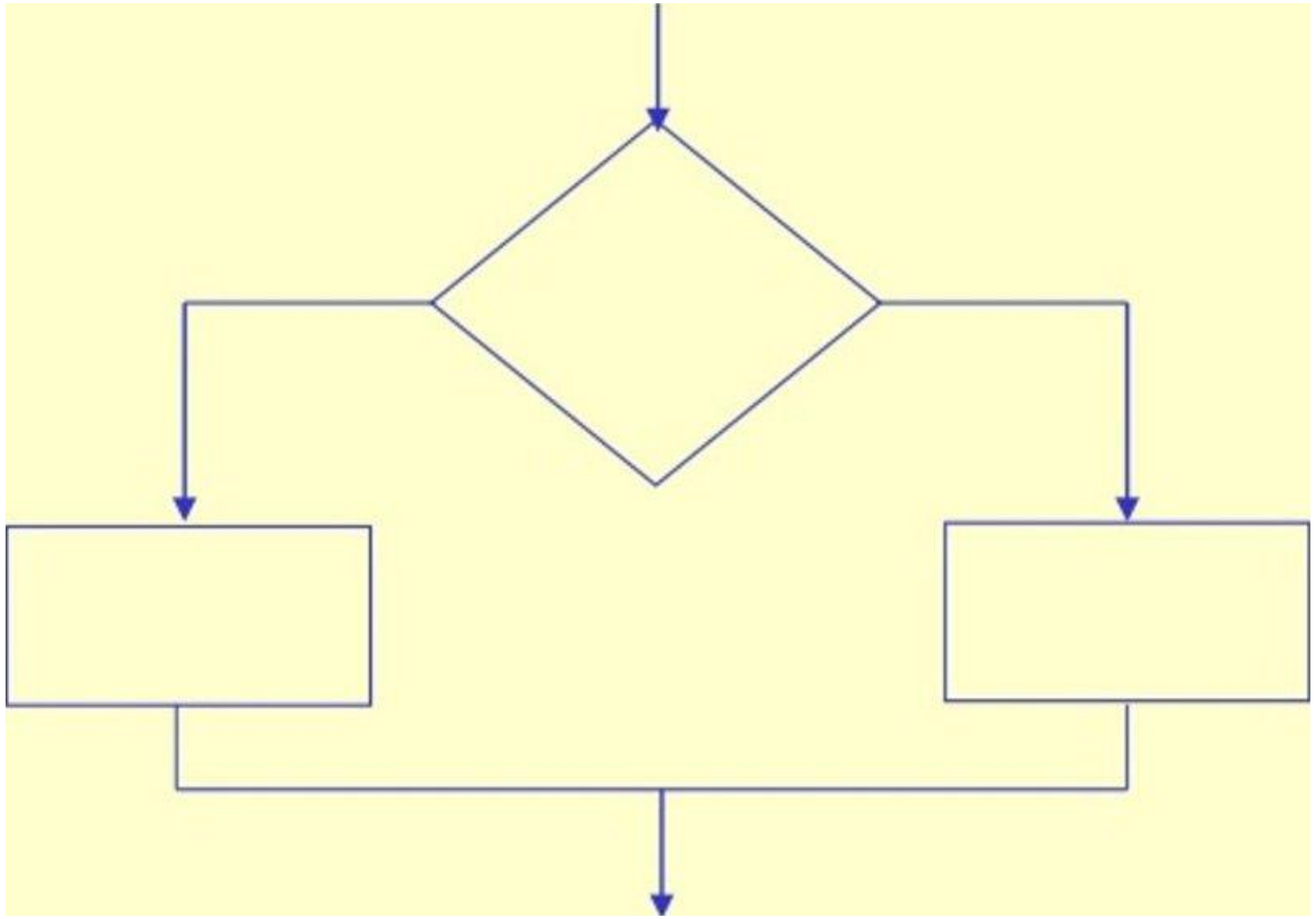
Review

- What type of structure is this?

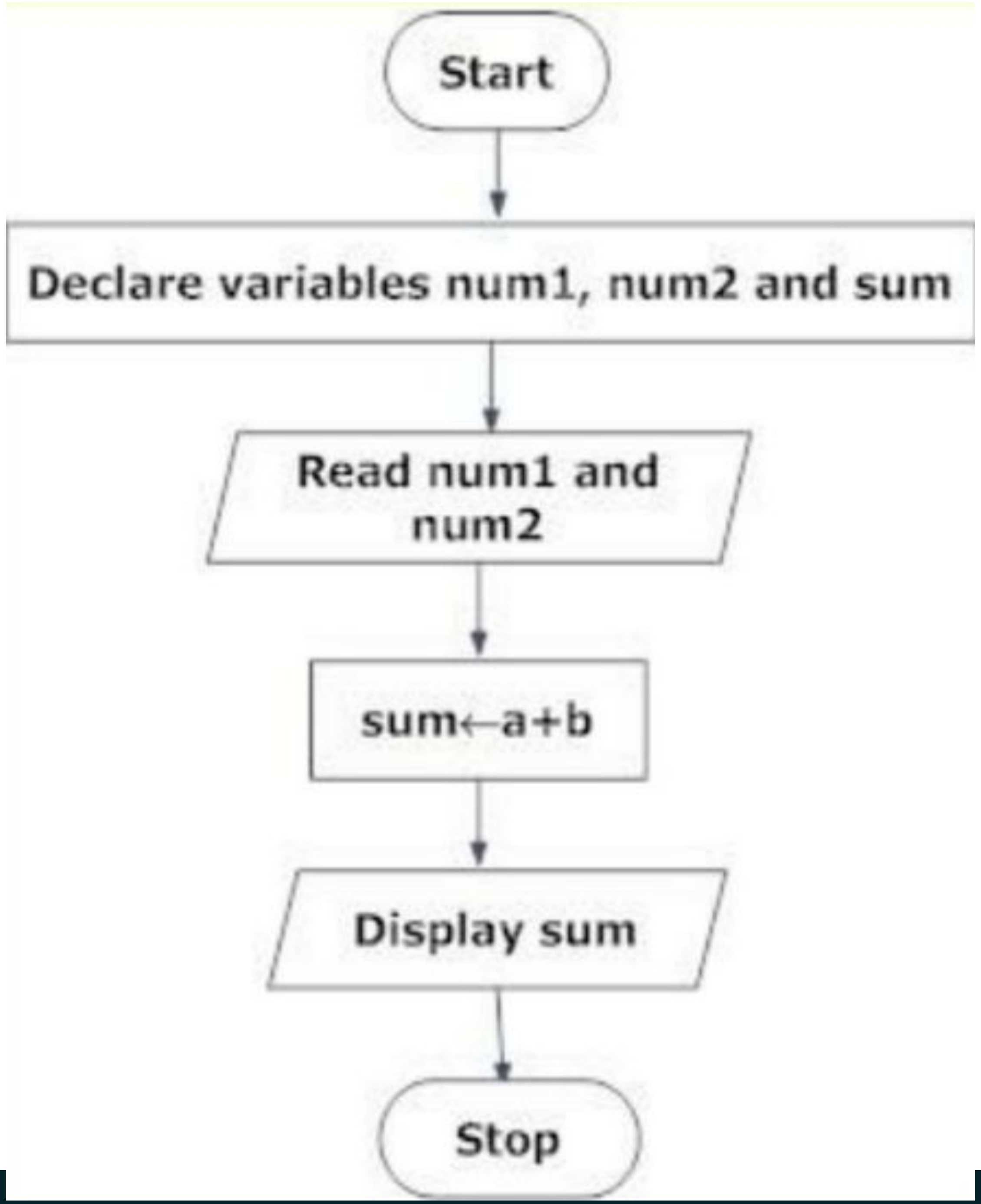


Answer

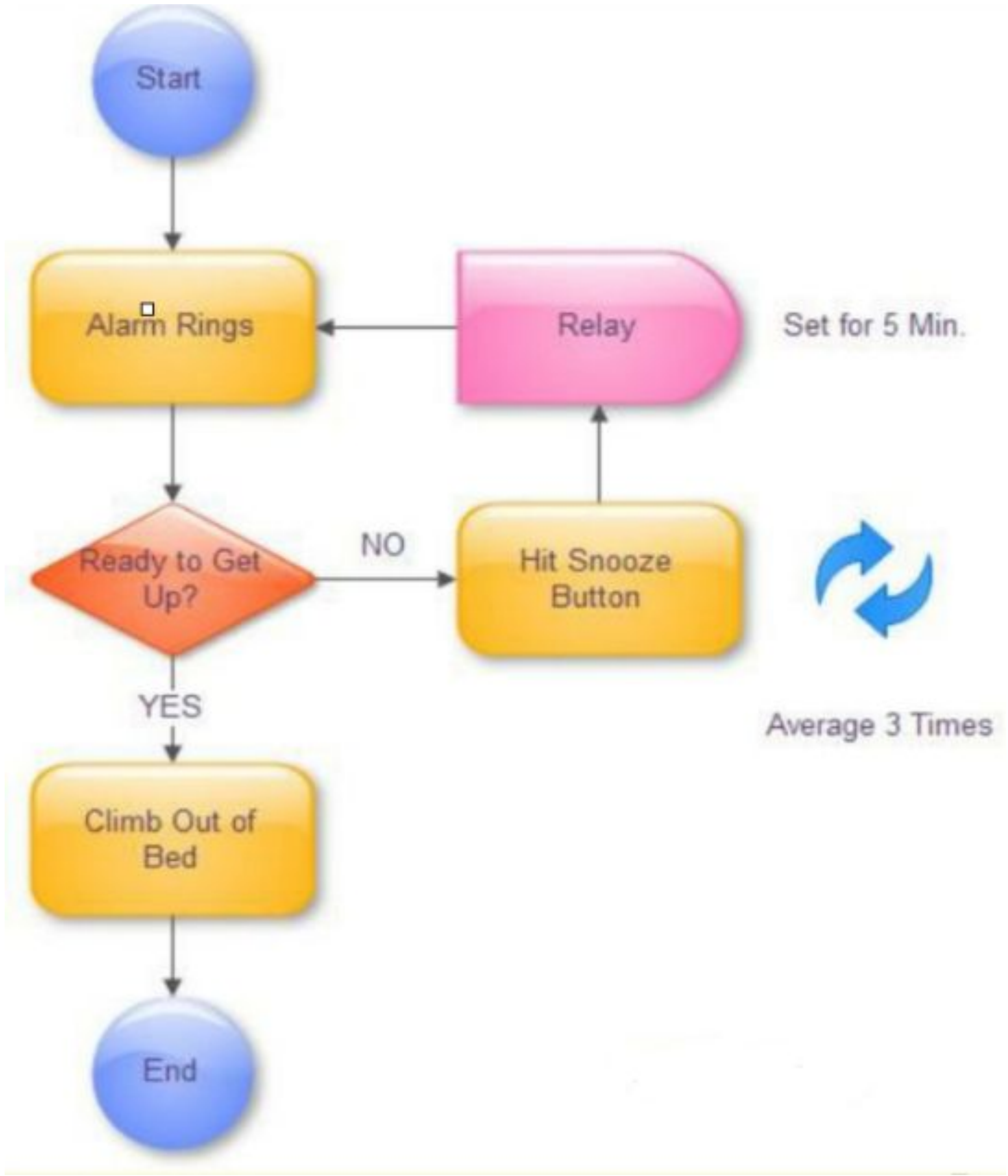
- Decision



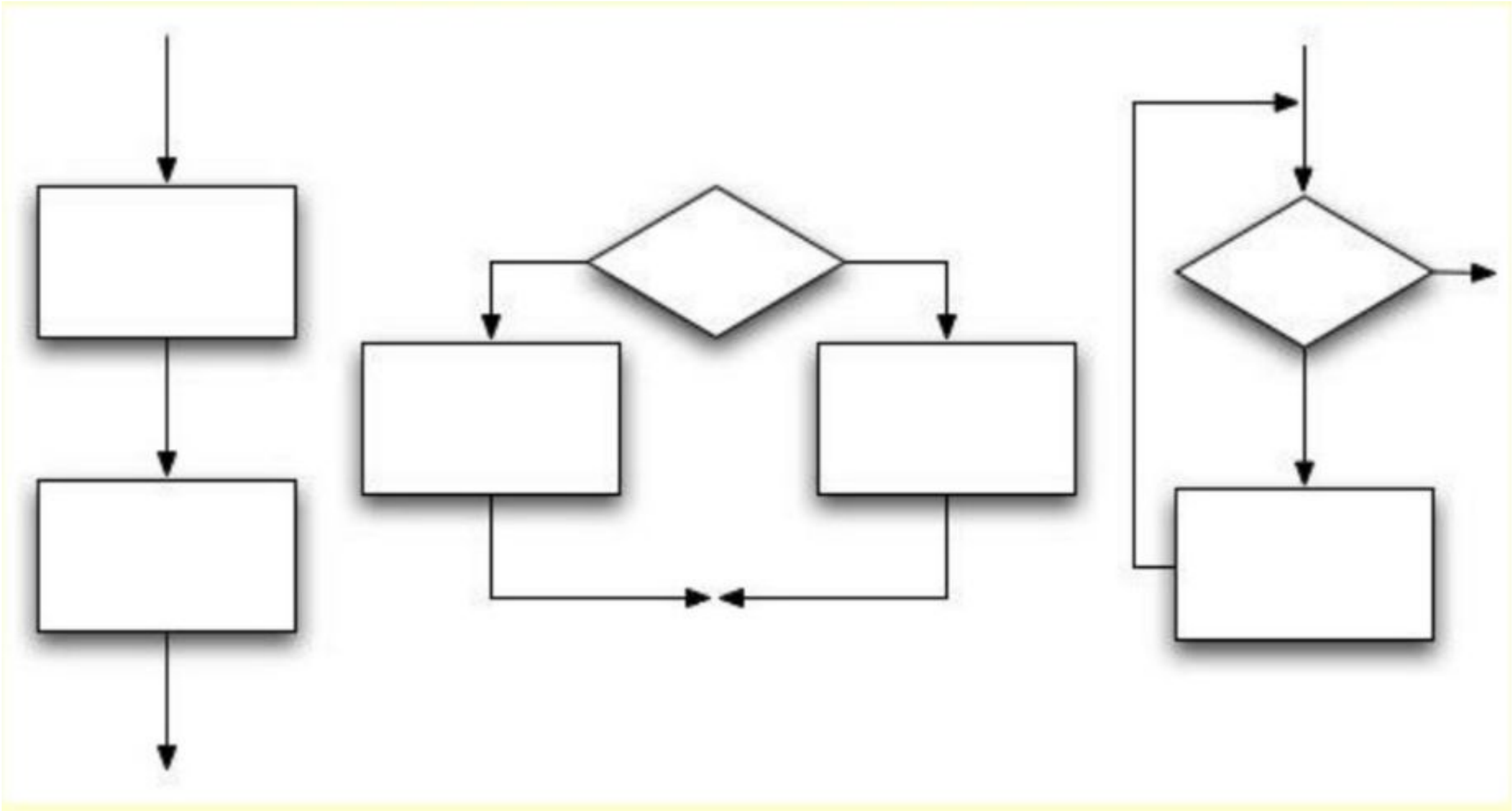
Example 1: Sum of two numbers



Example 2: Alaram Ringing



Examples????



Conclusion

Flowcharts are fundamental tools in computer science and engineering

- Essential for algorithm design and problem-solving
- Bridge the gap between problem and code
- Improve communication and documentation
- Foundation for more advanced concepts

Practice drawing flowcharts for every algorithm you learn!